

Quantum Fisher information as a thermal probe in frustrated magnets through insights from quantum spin ice

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Quantum Fisher information (QFI) is a measure of multipartite entanglement accessible via inelastic neutron scattering. Here we demonstrate that QFI reveals thermal and dynamical properties of quantum spin ice (QSI), a three-dimensional quantum spin liquid with fractionalized excitations. By developing a multi-directed loop update quantum Monte Carlo algorithm, along with exact diagonalization and gauge mean-field theory, we compute the QFI for the pyrochlore lattice. The temperature and momentum dependence of QFI maps the phase diagram, distinguishing the ferromagnetic ordered phase, its critical region, the zero-flux QSI, and the π -flux QSI. QFI also captures two crossover scales: from trivial paramagnet to classical spin ice, then to QSI. We discuss the π -flux QSI in light of experiments on cerium-based pyrochlores. Our results suggest that QFI not only detects entanglement but also serves as a sensitive thermal and dynamical probe for frustrated quantum magnets.

Entanglement is arguably the most important concept in quantum physics. It is at the heart of our modern understanding of the phases of matter and their transitions. As a notable example, entanglement is the fundamental characteristic of quantum spin liquids (QSLs)^{1–4}—paramagnetic phases of frustrated spin systems that fail to magnetically order down to zero temperature, and host fractional excitations and emergent gauge fields^{5–9}. QSLs are characterized by long-range entanglement, which may be measured for gapped QSL by a non-vanishing topological entanglement entropy^{4,10–13}. However, despite its role as a theoretical cornerstone and its usefulness in numerical studies^{12–17}, topological entanglement entropy is not experimentally accessible in solid-state platforms, where only local correlations are typically measured.

One still has experimental access to other measures of entanglement. A particularly useful one that has recently been at the forefront of the experimental search for QSLs is the quantum Fisher information (QFI)^{18–24}. Initially employed to define the maximal achievable precision in parameter estimation for a given quantum

state in the quantum metrology community^{25–30}, the QFI is directly related to the dynamical susceptibility³¹, which is routinely measured in inelastic neutron scattering experiments^{32–34}. The QFI density f_Q provides a lower bound on the multipartite entanglement in the system^{35,36}. This bound can be used to experimentally differentiate between QSLs and other, more trivial states, such as random singlet states driven by strong disorder^{24,37}. These states may otherwise be challenging to distinguish experimentally, as they can both yield similar signatures, such as continua of excitations in inelastic neutron scattering or the absence of experimentally observable finite-temperature phase transitions^{38–44}. Nevertheless, it should be emphasized that a large value of the QFI, although promising, does not provide evidence for the realization of a QSL in and of itself. Indeed, trivial ordered states sufficiently close to a quantum critical point may have an arbitrarily large value of f_Q ³¹. Be as it may, if precise theoretical predictions for the momentum and temperature dependence of the QFI for prospective QSLs exist, they offer stringent quantitative predictions which, if measured, may provide

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significantly more convincing evidence than qualitative features, like the presence of broad continua of excitations.

It is worth noting that the QFI is not meant to be a universal entanglement classifier; rather, it provides an operator-resolved, experimentally aligned witness for the entanglement depth of multipartite entanglement. Compared with other witnesses, QFI is distinguished in collective spin systems because it is simultaneously (i) well-defined for mixed thermal states, (ii) scalable to large many-body settings without requiring reduced density matrices, and (iii) directly linked to the excitation spectrum through a frequency integral of the dynamical response. By contrast, pairwise measures, such as concurrence^{37,45} probe only two-spin entanglement; they neither certify multipartite entanglement nor remain informative when entanglement is predominantly multipartite and delocalized, as is typical in collective correlated phases.

In this letter, we make such detailed predictions for the QFI on one of the most paradigmatic QSL: quantum spin ice (QSI)^{46–51}. QSI is a three-dimensional QSL that is the ground state of an XXZ model with dominant Ising and subleading transverse couplings on the pyrochlore lattice. It realizes the deconfined (Coulomb) phase of compact $U(1)$ gauge theory and, as such, hosts emergent photon excitations, spin-1/2 spinons that act as emergent electric charges, and magnetic monopoles⁵². QSI is the ideal platform for making specific predictions for the QFI, as it is one of the few known examples of an experimentally relevant model that stabilizes a well-understood QSL, which is numerically accessible over a parameter regime (i.e., ferromagnetic transverse couplings), with sign-problem-free quantum Monte Carlo (QMC)^{52–57}. Furthermore, several compounds have historically been considered as possible experimental realization of QSI, such as $\text{ Tb}_2\text{Ti}_2\text{O}_7$ ^{58–62}, $\text{ Pr}_2(\text{Sn,Zr,Hf})_2\text{O}_7$ ^{56,63–67}, and $\text{ Yb}_2\text{Ti}_2\text{O}_7$ ^{50,68–73}. The most recently considered candidate materials are the Cerium-based pyrochlore compounds $\text{ Ce}_2(\text{Zr,Hf,Sn})_2\text{O}_7$ ^{74–84}. For this last family of compounds, no magnetic order has been reported in $\text{ Ce}_2(\text{Zr,Hf})_2\text{O}_7$. Experimental determinations of their microscopic couplings indicate that $\text{ Ce}_2\text{Zr}_2\text{O}_7$ and $\text{ Ce}_2\text{Hf}_2\text{O}_7$ likely fall in a region of parameter space that stabilizes, so-called, π -flux QSI^{77,82–87}, where a constant flux of the emergent gauge field is threading hexagonal plaquettes, such that translation acts projectively on spinon excitations^{88–94}. Energy-integrated and inelastic neutron-scattering measurements consistent with theoretical predictions have also been reported^{77,79,83,86,94,95}, and a cubic scaling of the low-temperature heat capacity has even been recently measured in $\text{ Ce}_2\text{Zr}_2\text{O}_7$ ⁷⁹. Precise theoretical predictions for the QFI should thus be experimentally measurable and provide rigorous tests for these and future candidate materials.

We employ large-scale QMC simulations^{96–99} to evaluate the QFI of the XXZ model with ferromagnetic transverse coupling. To this end, we develop a multi-directed loop (MDL) update algorithm to efficiently sample the highly frustrated (3+1)d configurational space; the details are provided in the Supplemental Material (SM) (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the exact diagonalization (ED) and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.). ED is also used to compute the QFI^{100,101} for both antiferromagnetic and ferromagnetic transverse couplings. We show that ED and QMC results are consistent for ferromagnetic XY exchange, which justifies extending our ED result to the anti-ferromagnetic exchange case, where π -flux QSI is the ground state. These numerical results are further compared with predictions from the gauge mean-field theory (GMFT) parton construction^{88,89,93,94,102–105}, which provides a mapping from the spin Hamiltonian to a lattice $U(1)$ gauge theory coupled to bosonic matter fields in the low-temperature limit.

We find that the QFI at finite temperature is sensitive to thermal phase transitions and crossovers, as well as regions of large critical fluctuations, depending on the momentum position. For example, the QFI associated with the transverse spin components $S^\pm$ at the

$\Gamma = (0, 0, 0)$ point clearly maps out the thermal phase diagram of the pyrochlore XXZ model, revealing the two crossover temperature scales between the high-temperature paramagnet and classical spin ice as well as between classical spin ice and QSI^{52,56}. Its evolution also delineates the ferromagnetic (FM) thermal phase transition for large ferromagnetic transverse couplings^{53,55}. In contrast, the QFI at the $\Gamma' = (4\pi, 4\pi, 0)$ point appears to be sensitive to critical fluctuations near the phase transition between QSI and the ferromagnetic ordered phases. Altogether, our work provides the first large-scale, unbiased computation of experimentally accessible entanglement properties of QSI, offering exhaustive numerical results that can be quantitatively compared with future experimental results on current candidate materials.

Results Model

The pyrochlore lattice is a face-centered cubic lattice with four sublattices per unit cell, forming a network of corner-sharing tetrahedra. We consider spin-1/2 on the pyrochlore lattice that are interacting through an XXZ model of the form

$$H = J_z \sum_{\langle i,j \rangle} S_i^z S_j^z - J_\perp \sum_{\langle i,j \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad (1)$$

where the summation $\langle i,j \rangle$ runs over all the nearest-neighbor pairs. We focus on the regime with a dominant antiferromagnetic Ising coupling (i.e., $J_z > |J_\perp| > 0$), which provides geometric frustration within each tetrahedron and energetically favors “classical spin ice configurations” where $\sum_{i \in \text{tet}} S_i^z = 0$ with the sum over all spins in a given tetrahedron. In the perturbative regime $|J_\perp| \ll J_z$, the transverse coupling allows for tunneling between the classically degenerate configurations that respect this local energetic constraint⁴⁶. This tunneling stabilizes 0-flux QSI (QSI₀) for $0 < J_\perp/J_z < 0.052$ at low temperatures ($T \lesssim 12|J_\perp^3/J_z^2|$) as shown in QMC^{52–56}. When $-1 < J_\perp/J_z < 0$, π -flux QSI (QSI _{π}) is instead stabilized, where the flux refers to the emergent static magnetic flux threading the hexagonal plaquettes of the pyrochlore lattice in the ground state^{46,88,91,93,94,106–108}. For $J_\perp/J_z > 0.052$, the system undergoes a fluctuation-induced first-order transition^{109,110} into an XY ferromagnetic (FM) ordered phase below a critical temperature T_c ^{53,55,56}. Hereafter, we set $J_z = 1$ as the unit of energy. For reference, the dominant exchange is usually slightly smaller than 1K in Cerium-based dipolar-octupolar candidates^{77,82–87}.

To quantify the entanglement properties of a given state, the QFI provides a lower bound on multipartite entanglement, also known as entanglement depth^{37,45,111}. For a collective generator $\mathcal{O} = \sum_{i=1}^N \mathcal{O}_i$ with local eigenvalue range $\Delta\lambda = \lambda_{\max} - \lambda_{\min}$, any m -producible state ρ obeys^{20,35,36,45,112}

$$\frac{F_Q[\rho, \mathcal{O}]}{N} \leq m(\Delta\lambda)^2. \quad (2)$$

Equivalently, with $f_Q(\mathcal{O}) := F_Q[\rho, \mathcal{O}]/N$, we can define the normalized QFI (nQFI) as the lower bound of entanglement depth, such that

$$\text{nQFI}(\mathcal{O}) := \frac{f_Q(\mathcal{O})}{(\Delta\lambda)^2} > m \Rightarrow \text{entanglement depth} \geq m+1, \quad (3)$$

For the operator $\mathcal{O} = S_{\mathbf{q}}^\alpha := \sum_i S_{\mathbf{R}_i}^\alpha e^{i\mathbf{q} \cdot \mathbf{R}_i}$, the QFI density is related to the dynamical structure factor $A^\alpha(\mathbf{q}, \omega) := \frac{1}{2\pi N} \int dt \langle S_{\mathbf{q}}^\alpha(t) S_{\mathbf{q}}^\alpha(0) \rangle e^{i\omega t}$ at momentum $\mathbf{q}$ by³¹

$$f_Q(S_{\mathbf{q}}^\alpha, T) = 4 \int_0^\infty d\omega \tanh\left(\frac{\omega}{2T}\right) \left(1 - e^{-\omega/T}\right) A^\alpha(\mathbf{q}, \omega). \quad (4)$$

Here, T is the temperature and α labels different pseudospin components, such as $\alpha \in \{x, y, z\}$. We see that the QFI density $f_Q(S_{\mathbf{q}}^{\alpha}, T)$ can be obtained by integrating $A^{\alpha}(\mathbf{q}, \omega)$ over all frequencies at a fixed temperature and momentum. Experimental measurements of the QFI have already been reported for quasi-1D KCuF_2 ²⁰ and for 2D frustrated triangular lattice material KYbSe_2 ¹¹³.

To simulate the QSI_0 phase, we developed an MDL-QMC update that enables efficient simulations of the QSI_0 phase on the pyrochlore lattice. This task is difficult to achieve with the conventional directed loop update algorithm, as in the QSI_0 phase, the simplest quantum fluctuation that connects different classical spin-ice states involves six spin flips around a hexagon, which is hard to realize by a simple directed loop update. Our MDL algorithm overcomes this limitation by allowing the insertion of multiple operator pairs, thereby naturally generating higher-order processes during the Monte Carlo update and making the QSI_0 phase accessible (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation and the computation of neutron scattering cross section in an experimental setting.)). We then measure the imaginary time correlation function $G^{\pm} = \frac{1}{2N} \sum_{\gamma, \nu} \langle S_{\mathbf{q}, \nu}^{\pm}(\tau) S_{\mathbf{q}, \nu}^{\pm}(0) + S_{\mathbf{q}, \nu}^{\mp}(\tau) S_{\mathbf{q}, \nu}^{\mp}(0) \rangle$ and $G^z = \frac{1}{N} \sum_{\gamma, \nu} \langle S_{\mathbf{q}, \nu}^z(\tau) S_{\mathbf{q}, \nu}^z(0) \rangle$, where γ, ν label the four pyrochlore sublattices, the momentum-transfer $\mathbf{q}$ is measured in the 3D pyrochlore Brillouin zone (BZ), and $\tau \in [0, \beta]$ denotes the imaginary time with $\beta = 1/T$ the inverse temperature. We utilize the stochastic analytic continuation (SAC) scheme^{114–118} to convert the imaginary time correlation function to a real-frequency dynamic structure factor. This QMC+SAC scheme has been successfully applied to a variety of lattice models, producing reliable spectral properties ranging from magnon and amplitude modes in a magnetically ordered state^{119,120} to fractionally excited excitations in QSL and QSI models^{52,121–123}.

We further compute the QFI density using ED and GMFT. The ED calculations are performed over a 16-site periodic cubic conventional cluster (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering

cross section in experimental setting.)), providing access to momentum positions Γ , Γ' , and X . We first compare the ED and GMFT results with our QMC data before extending these calculations into the π -flux $J_{\pm} < 0$ regime, where most QSI candidate materials reside and QMC encounters a sign problem.

Numerical results

Our main results are summarized in Fig. 1. The white dashed line marks the phase boundary between the paramagnetic and FM phases, whereas the white solid line ($J_{\pm, c} = 0.052$) indicates the transition from the FM to the QSI_0 phases in panel (b). Let us first define the total spinon QFI $f_Q(S_{\mathbf{q}}^{\pm}, T) = f_Q(S_{\mathbf{q}}^x, T) + f_Q(S_{\mathbf{q}}^y, T)$ under the XXZ model. Panels (a, b) and (c, d) show $f_Q(S_{\mathbf{q}}^{\pm}, T)$ and $f_Q(S_{\Gamma'}^{\pm}, T)$, respectively, at different values of $J_{\pm}$. Among which, (a, c) are obtained from ED and (b, d) are from QMC on a $L = 4$ lattice ($4 \times L^3$ sites). We additionally computed the QFI obtained from GMFT and compared it with QMC simulations at $J_{\pm} = 0.045$ for system sizes $L = 3$ and $L = 4$. We find that, while QMC and ED results are consistent with each other, GMFT results are only off by a factor of approximately 6/7 compared to QMC and ED at low temperatures (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.)). This is impressive given the mean-field nature of the GMFT. As such, we argue that the ED and GMFT results can be readily extended into the $J_{\pm} < 0$ regime.

We would like to first highlight that QFI shows non-trivial entanglement depth across various phases. For spin-1/2 systems, we find $n\text{QFI}(S_{\mathbf{q}}^{\pm}) = f_Q(S_{\mathbf{q}}^{\pm}, T)/2$ (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.)). As shown in Fig. 1a, b, $f_Q(S_{\Gamma'}^{\pm}, T) \geq 3$ in QSI_0 and ≥ 5 in the FM phase, implying at least 2- and 3-partite entanglement, respectively, in sharp contrast to the high-temperature paramagnet (PM), where $f_Q(S_{\Gamma'}^{\pm}, T) = 0$. On the other hand, we see that $f_Q(S_{\Gamma}^{\pm}, T)$ of QSI_{π} is ~ 0 . We hereby stress that QFI is a

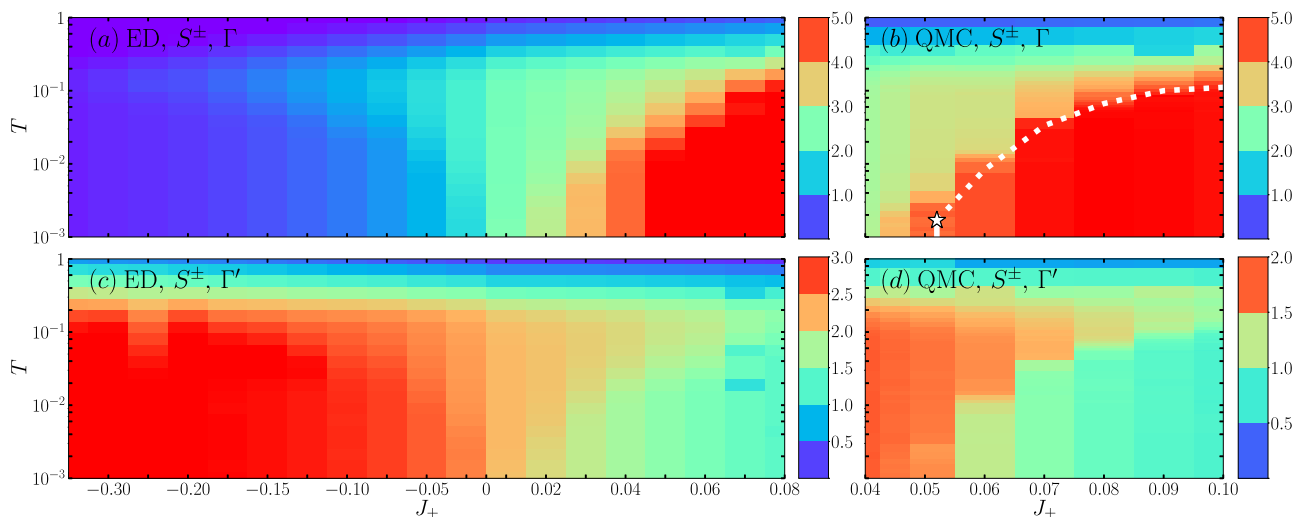


Fig. 1 | Heat maps of the QFI as functions of temperature T and $J_{\pm}$. **a, b** show the QFI density $f_Q(S_{\mathbf{q}}^{\pm}, T)$ in the S^z channel at $\Gamma = (0, 0, 0)$, while **c, d** show it at $\Gamma' = (4\pi, 4\pi, 0)$. **a, c** are obtained from the ED calculation of 16-site cluster with $J_{\pm}$ ranging from -0.045 to 0.08 , and **b, d** are from the QMC simulation of $4 \times L^3$ sizes ($L = 4$) with $J_{\pm}$ ranging from 0.04 to 0.10 . In **(a)** and **(b)**, $f_Q(S_{\Gamma}^{\pm}, T)$ maps out the thermodynamic phase boundaries between the ferromagnetic phase (FM) and QSI_0 . The temperature dependence of $f_Q(S_{\Gamma'}^{\pm}, T)$ from QMC further discerns the crossover temperature scales between the high-temperature paramagnetic regime to classical spin ice and eventually to the QSI_0 regime (see also Fig. 2a). In panels (c)

and **(d)**, $f_Q(S_{\Gamma'}^{\pm}, T)$ reflects the strength of the fluctuations in the thermal and quantum phase diagram, with strong QFI at the classical critical region above the FM phase and stronger QFI in the classical spin ice and QSI_0 region. Moreover, the strongest QFI signal, represented by the red region in **(c)** for $J_{\pm} < 0$, reflects the QSI_{π} regime (see also Fig. 2b). In **(b)**, the white dashed line denotes the phase boundary between the paramagnetic and FM phases, whereas the white solid line ($J_{\pm, c} = 0.052$) represents the transition between the FM and QSI_0 phases. The white star marks the crossover temperature scale from the CSI regime to the QSI_0 regime ($T = 12 \times 0.052^3 \approx 0.00169$) at $J_{\pm, c}$.

mere lower bound of entanglement depth, such results do not reflect that QSI_π is trivially entangled nor that the FM phase is more entangled than that of the QSI_π phase. In fact, QFI is heavily dependent on the momentum positions. In principle, to aptly categorize the lower bound of the entanglement depth of a certain phase, one should rigorously search through the momentum space to find the largest QFI. One guiding principle for determining said momentum position is to examine the equal-time spin structure factor (ETSF), since the QFI density at $T \rightarrow 0$ becomes $\langle S_{\mathbf{q}}^+ S_{\mathbf{q}}^- + S_{-\mathbf{q}}^+ S_{\mathbf{q}}^+ \rangle / 2$. We demonstrate this point by presenting $f_Q(S_{\mathbf{r}}^{\pm}, T)$ in Fig. 1c, d, where $\Gamma' = (4\pi, 4\pi, 0)$, at different values of $J_{\pm}$. This momentum position corresponds to a local maximum of the ETSF for QSI_π^{93,94} and, consequently, a local maximum for QFI at low temperatures^{20,45}. Indeed, here $f_Q(S_{\mathbf{r}}^{\pm}, T)$ of QSI_π is ≥ 3 , whereas for QSI₀ and FM they are ≥ 2 and ≥ 1 , respectively. From this, we can conclude that QSI_π is at least 2-partite entangled.

Beyond providing a lower bound on entanglement depth, the QFI is a channel-resolved measure of quantum coherence, and therefore depends explicitly on the choice of probe operator. For thermal states, the temperature kernel in Eq. (4) effectively suppresses quasi-static contributions with $\omega \ll T$, which are strongly thermally populated and typically correspond to slow, relaxational fluctuations that dominate classical noise in equal-time correlators and static susceptibilities^{23,31,124}. In contrast, the QFI preferentially weights coherent dynamical fluctuations at $\omega \gtrsim T$ in the specific channel accessed by $S_{\mathbf{q}}^{\alpha}$, so changing α and $\mathbf{q}$ directly tunes which excitations contribute most strongly^{23,31,125}. This built-in quantum selectivity and operator tunability make f_Q a sharp diagnostic of the thermal crossovers and phase boundaries. In particular, at finite temperature, $f_Q(S_{\mathbf{r}}^{\pm}, T)$ faithfully tracks the two crossover temperatures of QSI₀ regime as shown in Fig. 2a for $J_{\pm} = 0.045$ and 0.05 . From previous QMC calculations of the specific heat⁵², it is known that as one cools down from the high-temperature paramagnetic (PM) phase, the system first crosses over into the classical spin ice (CSI) at $T \sim 1$ before finally entering the QSI₀ regime at $T \sim |J_{\pm}^3|$. These crossover temperatures are marked by the two peaks in the specific heat. But these features are equally well represented in the QFI density. In Fig. 2a, $f_Q(S_{\mathbf{r}}^{\pm}, T)$ first starts increasing from 0 in the PM phase at $T \sim 1$ when we cross over into CSI. After which, $f_Q(S_{\mathbf{r}}^{\pm}, T)$ plateaus until the second temperature scales $T \sim |J_{\pm}^3|$ where $f_Q(S_{\mathbf{r}}^{\pm}, T)$ rises again as we cross over into the QSI₀ phase. Finally, $f_Q(S_{\mathbf{r}}^{\pm}, T)$ reaches another plateau, which is consistent with the GMFT calculation (the green triangle in Fig. 2a) up to a factor of approximately $\sim 6/7$. Similarly, the phase boundary between the FM phase and the high-temperature PM is clearly delineated by the increase in $f_Q(S_{\mathbf{r}}^{\pm}, T)$, as shown in Fig. 2a. As such, the QFI also appears to be sensitive to this thermal phase transition⁵³.

On the other hand, the two temperature crossover scales are also clearly demarcated by QFI for QSI_π phase, as shown in Fig. 1a, c: both $f_Q(S_{\mathbf{r}}^{\pm}, T)$ and $f_Q(S_{\mathbf{r}}^{\pm}, T)$ increase near $T \sim 1$; at $T \sim |J_{\pm}^3|$, $f_Q(S_{\mathbf{r}}^{\pm}, T)$ decreases while $f_Q(S_{\mathbf{r}}^{\pm}, T)$ increases for $J_{\pm} \lesssim -0.3$. At $J_{\pm} \sim -0.3$, the two temperature scales coalesce, as shown by the essentially flat QFI below the first temperature scale 1K in Fig. 2a, b. The behavior is consistent with Ce₂Zr₂O₇ (best-fit $J_{\pm} \approx -0.3$ ^{77,80,86}), where only one peak is observed in specific heat measurement⁸⁶. In contrast, specific heat measurements on Ce₂Hf₂O₇ ($J_{\pm} \approx -0.125$ in the QSI scenario)^{82–84,125} show two well-separated peaks⁸³ mirrored by the predicted two inflection points of QFI, most pronounced in $f_Q(S_{\mathbf{r}}^{\pm}, T)$ as shown in Fig. 2a.

The evolution of QFI also strongly depends on the momentum position. The two aforementioned temperature scales manifest themselves in a strikingly different way when looking at $f_Q(S_{\mathbf{r}}^{\pm}, T)$ in Fig. 2b as opposed to $f_Q(S_{\mathbf{r}}^{\pm}, T)$ in Fig. 2a. As we cool down to the CSI regime from the high-temperature PM at the temperature scale J_z , $f_Q(S_{\mathbf{r}}^{\pm}, T)$ rapidly rises and eventually plateaus. But after the second temperature scale is met, $f_Q(S_{\mathbf{r}}^{\pm}, T)$ now dips instead of increasing, giving rise to this characteristic hump as shown in Fig. 2b, which

precisely marks the region of large critical fluctuations. This speaks to the versatility of QFI as a potentially powerful experimental probe—by adjusting the momentum positions, we can design probes that are sensitive to specific features of interest.

Discussion

So far, we have considered the QFI of an abstract XXZ model. The same framework connects directly to real neutron scattering experiments once the microscopic nature of the local moments and their coupling to the neutron's magnetic moment are specified (see detailed derivation in SM. (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.)). In particular, we present predictions in Fig. 3 for the Ce-based dipolar-octupolar pyrochlore Ce₂Zr₂O₇^{77,86,126}—a leading QSI candidate. While Ce ions carry both dipolar and octupolar moments, neutron scattering primarily couples to the dipolar moments, which correspond to transverse spin components $S^{\pm}$ in the current model^{74,127}. Figure 3 shows the corresponding QFI $f(S_{\mathbf{q}}^{\text{DO}}, T)$ that takes into account the transverse momentum projector in the inelastic neutron scattering.

In Fig. 3a, b, re obtained from QMC simulation for $L = 4$ and panels (c, d) are the ED results. One sees that in the global frame, the QFI again vividly captures the thermodynamic phase structure. The crossover temperature scale from paramagnetic region at high temperature ($T > J_z$) to the CSI and its plateau at intermediate temperature ($|J_{\pm}^3| < T < J_z$) and eventually the QSI₀ at low temperature ($T < |J_{\pm}^3|$) manifest in the both the ED and QMC data. The evolution of QFI closely resembles that of the thermal entropy measurements⁵².

One important point is that, in principle, the bounds on $f_Q(S_{\mathbf{q}}^{\text{DO}}, T)$ for an m -producible state are momentum dependent because of the transverse projector. But we can establish a $\mathbf{q}$ -independent bound as derived in SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.), giving us the following relationship: $n\text{QFI}(S_{\mathbf{q}}^{\text{DO}}) = \frac{3}{2}f_Q(S_{\mathbf{q}}^{\text{DO}})$. As a result, values extracted directly from our computed data in a neutron scattering setting already certify nontrivial entanglement depth: $n\text{QFI}(S_{\mathbf{r}}^{\text{DO}}, T) \geq 2.2$ for 0-flux QSI and $n\text{QFI}(S_{\mathbf{r}}^{\text{DO}}, T) \geq 3.7$ for all-in-all-out order (FM in the local spin frame), respectively. At the $X = (0, 0, 2\pi)$ point, $n\text{QFI}(S_X^{\text{DO}}, T) \geq 2.3$ certifies that the π -flux QSI is at least 3-partite entangled. Moreover, in principle, polarized neutron measurements with the non-spin-flip (NSF) and spin-flip (SF) channels are the optimal probes to find the best entanglement depth bound. As shown in SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.), $n\text{QFI}(S_{\mathbf{q}}^{\text{(NSF)}}) = 3f_Q(S_{\mathbf{q}}^{\text{(NSF)}})$, therefore, even a modest $f_Q(S_{\mathbf{q}}^{\text{(NSF)}})$ can signal deep multipartite entanglement.

Based on the results and analysis above, we believe the QFI can actually offer richer information than initially anticipated. Besides providing insights into the entanglement depth, it can serve as both a thermal and a dynamical probe in the investigation of quantum magnets, with quantum spin ice as an archetypal manifestation. Based on spin spectra in both local and global frames, the temperature- and momentum-dependent QFI faithfully traces the ground-state and finite-temperature phase boundaries in the QSI system, consistent with previous theoretical and numerical results. These properties of QFI can bridge the fundamental concept of entanglement to practical model computations and experimental applications. In this way, other exotic quantum critical points, QSL phases, and their associated theory, computation, and experimental verification, could be expected to unite from the entanglement perspective.

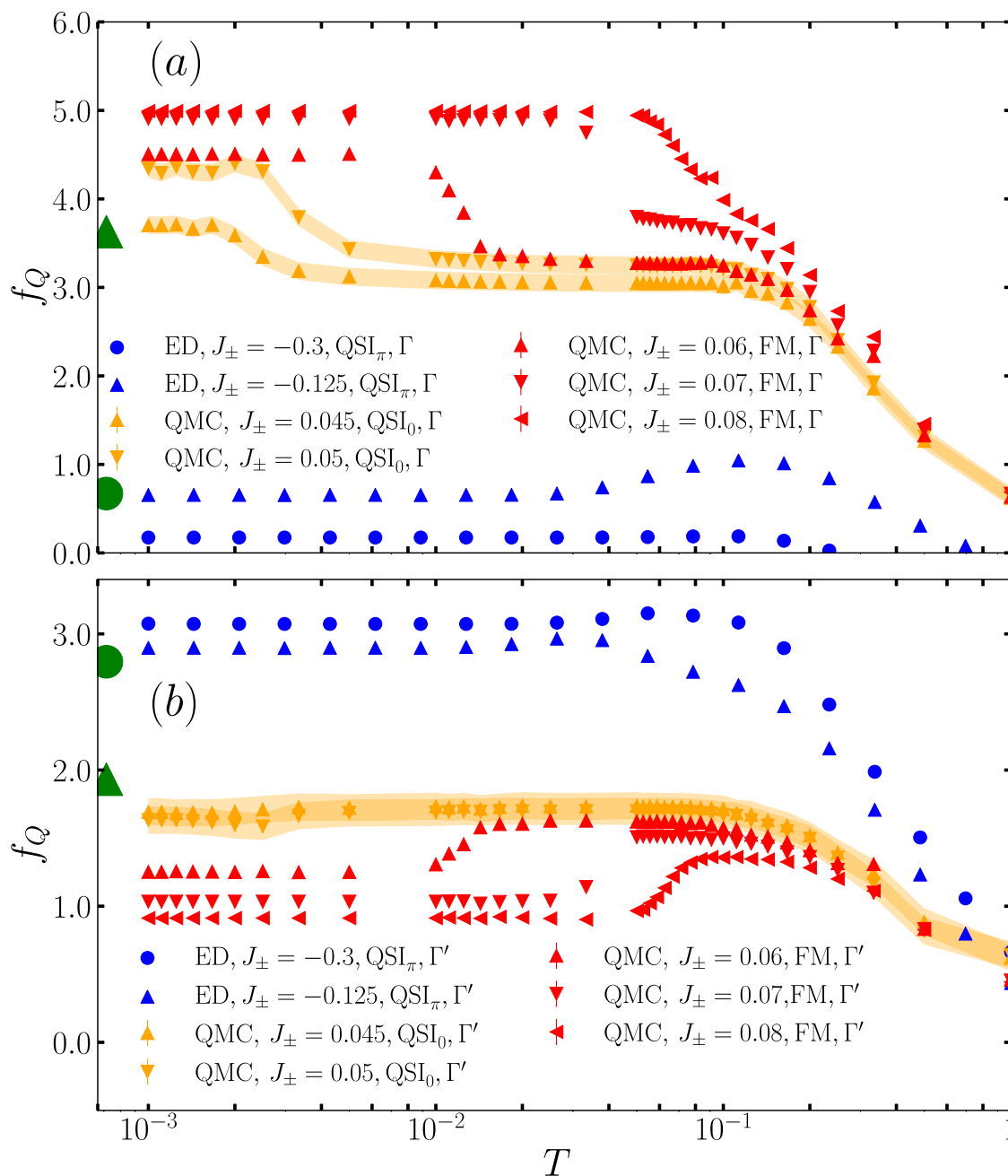


Fig. 2 | Temperature evolution of the QFI in different phases obtained with various computational methods. The QFI $f_Q(S_q^z, T)$ are shown for (a) the S^z channel at the Γ point, (b) the S^z channel at the Γ' point. The blue disk points represent results from ED calculations with $J_{\pm} = -0.3$ and $J_{\pm} = -0.125$ in the QSI_{π} regime, while the orange triangle correspond to QMC simulations with $J_{\pm} = 0.045$ and $J_{\pm} = 0.05$ in the QSI_0 regime. The orange shaded areas accompanying the data highlight the crossover from paramagnetic regime to classical spin ice regime at $T \sim 1$ and that from the classical spin ice regime to QSI_0 at $T \sim |J_{\pm}^3|$. The red triangles

indicate QMC results with $J_{\pm}$ ranging from 0.06 to 0.08 in the FM regime. The green circles near the y-axis indicate the GMFT calculation results at $J_{\pm} = -0.3$ (QSI_{π}) at zero temperature, while the green triangle corresponds to the result at $J_{\pm} = 0.045$ (QSI_0). The QMC, ED, and GMFT results are consistent (see SM (The Supplemental Material details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the exact diagonalization and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.) for details).

Methods

The numerical results for the pyrochlore lattice XXZ model are obtained from three methods: QMC simulations with an MDL update scheme, ED, and gauge mean-field theory (GMFT) calculations. In addition, we transfer our observations to the neutron scattering section that connects directly to the real experiments. Here, we briefly introduce all three numerical methods and this transformation. The details of these methods are provided in the SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI

model, the ED and the gauge field theory calculation and the computation of neutron scattering cross section in an experimental setting.).

Quantum Monte Carlo simulations with the multi-directed loop update

The QMC simulations are performed based on the stochastic series expansion (SSE) framework^{96,97,99}, where the configuration space is constructed by expanding the partition function, $Z = \text{Tr}(e^{-\beta H})$, as a Taylor series into an operator string. We apply the diagonal update,

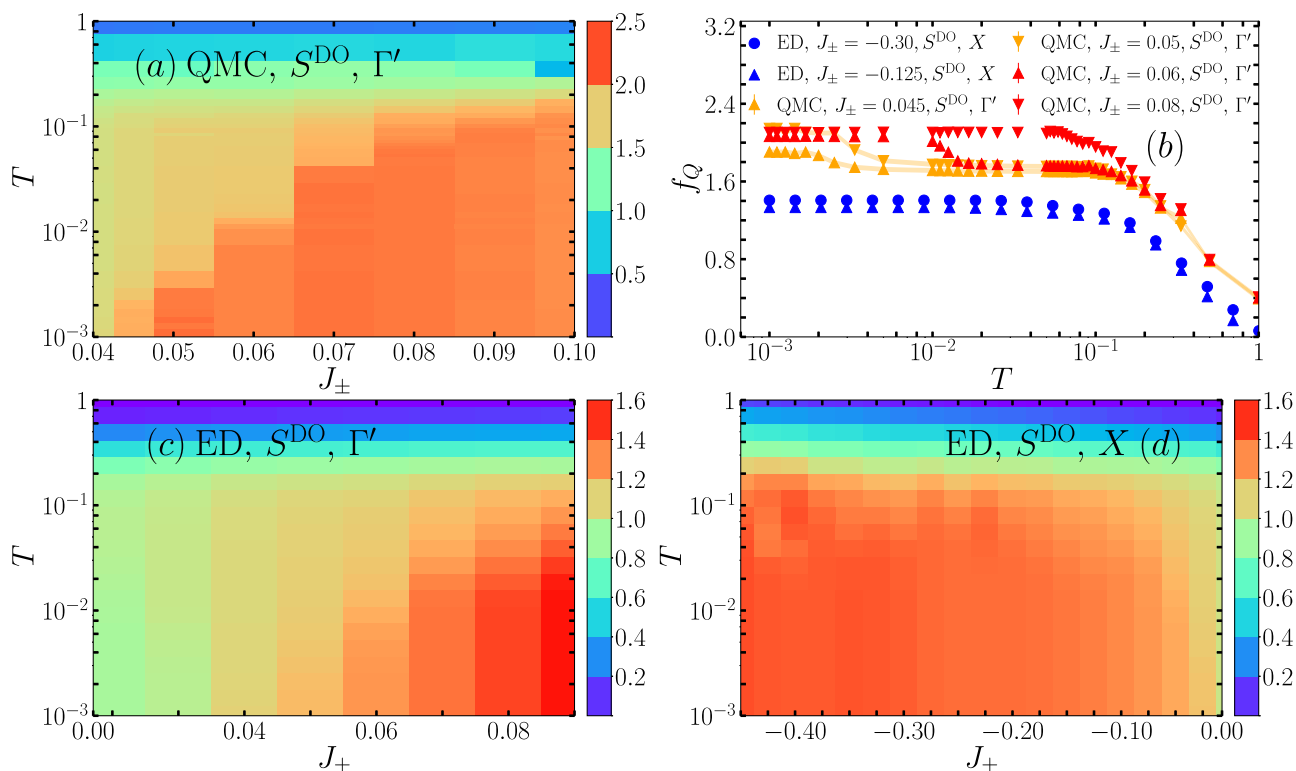


Fig. 3 | QFI in experimental coordinates for Cerium-based pyrochlore compounds. **a** presents a heat map of the QMC computed $f_Q(S_r^{\text{DO}}, T)$, with $J_{\pm}$ ranging from 0.04 to 0.10. **b** is the line cut of **(a)** from $J_{\pm} = 0.045$ to 0.08. The crossover temperature scales from the paramagnetic phase to CSI at $T \sim 1$ and from

CSI to QSI₀ at $T \sim |J_{\pm}^3|$ are clearly manifest in the shaded data, highlighted in orange. **c** shows ED results of $f_Q(S_r^{\text{DO}}, T)$ for $J_{\pm}$ between 0.01 and 0.08. **d** focuses $f_Q(S_X^{\text{DO}}, T)$ with $X = (0, 0, 2\pi)$ in the range from -0.425 to 0.00.

directed loop (DL) update, and MDL update schemes in our simulation. Our simulations are performed on system sizes $L = 3$ and 4. The MDL update is specifically designed to efficiently sample the highly frustrated (3+1)d configurational space of the pyrochlore lattice XXZ model, where the simplest quantum fluctuation that connects different classical spin-ice states involves six spin flips around a hexagon. This nature limits the sampling efficiency of the traditional DL update. To overcome this limitation, the MDL update allows the insertion of multiple operator pairs, thereby achieving efficient sampling and making the QSI₀ phase accessible (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation and the computation of neutron scattering cross section in experimental settings.) for details. In this update method, we first randomly select a hexagon in the configurational space and insert three pairs of operators, S_i^+ and S_i^- , on three alternate sites of the hexagon. Then, for each pair of operators, we randomly choose one operator to be the head of the loop, while the other operator becomes the tail. These three heads then propagate through the operator string until all of these heads meet their tails again at the same propagation step, at which point all loops are closed, and the update process is completed.

In addition, we measure the imaginary time correlation function by tracing the evolution of the configuration space in the QMC simulation¹²⁰. We then utilize the SAC scheme^{114–118} to convert the imaginary time correlation function to a real-frequency dynamic structure factor.

Exact diagonalization

The ED scheme follows that of the microcanonical thermal pure quantum method (mTPQ)^{100,101}. The spectral function is obtained from an mTPQ representative state $|\psi_k\rangle$ using the broadened resolvent of a

chosen operator $\mathcal{O}$ ^{128,129},

$$G_{\mathcal{O}}(z; \beta_k) = \frac{\langle \psi_k | \mathcal{O}^\dagger (z - \tilde{H})^{-1} \mathcal{O} | \psi_k \rangle}{\langle \psi_k | \psi_k \rangle}, \quad (5)$$

with $\tilde{H} \equiv H - E$ and $z = \omega + i\eta$. Here $\eta > 0$ controls the Lorentzian broadening, and the shift E fixes the absolute frequency reference. We choose E as the ground-state energy, obtained independently from a separate Lanczos run, so that the spectrum is reported with the correct energy offset. The associated spectral density follows as

$$I_{\mathcal{O}}(\omega; \beta_k) = -\frac{1}{\pi} \text{Im} G_{\mathcal{O}}(\omega + i\eta; \beta_k). \quad (6)$$

Specifically, we perform the ED calculations on a 16-site periodic lattice, where four tetrahedra construct a large tetrahedron. Such a geometry respects the full symmetry of the pyrochlore lattice.

Gauge mean-field theory calculations

The gauge mean-field theory (GMFT) calculations are performed to calculate the dynamical structure factor and QFI by canonically mapping the spin Hamiltonian to that of a lattice $U(1)$ gauge theory coupled to bosonic matter fields. We note that the results of GMFT align well with the QMC results with a factor of approximately 6/7 (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.) for details.

Neutron scattering cross section

The neutron scattering cross-section is proportional to

$$A^{\text{DO}}(\mathbf{q}, \omega) = \sum_{\mu, \nu} \left(\hat{\mathbf{z}}_{\mu} \cdot \hat{\mathbf{z}}_{\nu} - \frac{(\hat{\mathbf{z}}_{\mu} \cdot \mathbf{q})(\hat{\mathbf{z}}_{\nu} \cdot \mathbf{q})}{|\mathbf{q}|^2} \right) A_{\mu, \nu}^{\pm}(\mathbf{q}, \omega), \quad (7)$$

where μ and ν are the four sublattice indices of the pyrochlore lattice, and $A_{\mu, \nu}^{\pm}(\mathbf{Q}, \omega)$ is the spectral function obtained in the local frame. $\hat{\mathbf{z}}_{\mu}$ is the local z -axis direction of sublattice μ , which are given by

$$\begin{aligned} \mathbf{z}_0 &= \frac{1}{\sqrt{3}}(1, 1, 1) & \mathbf{z}_1 &= \frac{1}{\sqrt{3}}(1, -1, -1) \\ \mathbf{z}_2 &= \frac{1}{\sqrt{3}}(-1, 1, -1) & \mathbf{z}_3 &= \frac{1}{\sqrt{3}}(-1, -1, 1) \end{aligned} \quad (8)$$

Quantum Fisher information

A meaningful QFI bound is defined with respect to a particular collective observable $\mathcal{O} = \sum_i \mathcal{O}_i$, such that $f_Q(\mathcal{O}) \leq m(\Delta\lambda)^2$ when evaluated over an m -producible state, and $\Delta\lambda$ is the spectral width of the individual observables $\mathcal{O}_i$. However, in the main text, we use $S_{\mathbf{q}}^{\alpha} = \sum_i S_i^{\alpha} e^{i\mathbf{q} \cdot \mathbf{R}_i}$, which is not always Hermitian. Nevertheless, with the definition in Eq. (4), we now show how to extract the essential properties of the QFI f_Q , such as the entanglement lower bound. A detailed derivation is provided in the SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation and the computation of neutron scattering cross section in an experimental setting.).

For a given operator $\mathcal{O}_{\mathbf{q}} = \sum_i \mathcal{O}_i e^{i\mathbf{q} \cdot \mathbf{R}_i}$, we use the Hermitian $\mathcal{O}_{c, \mathbf{q}} = (\mathcal{O}_{\mathbf{q}} + \mathcal{O}_{\mathbf{q}}^{\dagger})/2$ and the antihermitian $\mathcal{O}_{s, \mathbf{q}} = (\mathcal{O}_{\mathbf{q}} - \mathcal{O}_{\mathbf{q}}^{\dagger})/2i$ parts to construct a QFI matrix (QFIM) $F^{130-132}$, whose elements are specified by:

$$F_{ab}(T) := 4 \int d\omega \tanh\left(\frac{\omega}{2T}\right) (1 - e^{-\omega/T}) \mathcal{A}_{ab}(\mathbf{q}, \omega), \quad (9)$$

where

$$\mathcal{A}_{ab}(\mathbf{q}, \omega) = \frac{1}{2\pi N} \int dt e^{i\omega t} \langle \mathcal{O}_{a, \mathbf{q}}(t) \mathcal{O}_{b, \mathbf{q}}(0) \rangle, \quad (10)$$

with $a, b \in \{c, s\}$. $f_Q(\mathcal{O}_{\mathbf{q}})$ defined in Eq. (4) is equivalent to $\text{Tr}(F)$ (see SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation, and the computation of neutron scattering cross section in experimental setting.) for more details). Therefore, since both diagonal elements are bounded by Eq. (2), for an m -producible state, $f_Q(\mathcal{O}_{\mathbf{q}}) = f_Q(\mathcal{O}_{c, \mathbf{q}}) + f_Q(\mathcal{O}_{s, \mathbf{q}})$ is simply bounded by the sum of the individual QFI bounds. After detailed derivation, we find that:

$$f_Q(\mathcal{O}_{\mathbf{q}}, T) \leq m(\Delta(\mathcal{O}_i))^2, \quad (11)$$

where $\Delta(\mathcal{O}_i)$ is the spectral width of the operator $\mathcal{O}_i$. When $\mathcal{O}_{\mathbf{q}} = S_{\mathbf{q}}^{\alpha}$ ($\alpha \in \{x, y, z\}$) and $S_{\mathbf{q}}^{\pm}$, the corresponding entanglement depth lower bound condition is that when

$$f_Q(S_{\mathbf{q}}^{\alpha}, T) > m, f_Q(S_{\mathbf{q}}^{\pm}, T) > 2m, \quad (12)$$

the state is at least $(m+1)$ -partite entangled.

Furthermore, we show how to extract essential properties of the QFI for the neutron scattering cross section via Eq. (7). This is noteworthy because there does not exist any collective operator, S^{DO} , such that its two-point correlation function, $\langle S^{\text{DO}} S^{\text{DO}} \rangle$, is equivalent to A^{DO} . Nevertheless, we can rewrite it as a sum of the QFI induced by the non-spin-flip (NSF) channel with underlying operator

$S_{\mathbf{q}}^{\text{NSF}} = \sum_{\mathbf{R}_{\mu}} (\hat{\mathbf{p}} \cdot \hat{\mathbf{z}}_{\mu}) \tau_{\mathbf{R}_{\mu}}^z e^{i\mathbf{q} \cdot \mathbf{R}_{\mu}}$ and the spin-flip (SF) channel with operator $S_{\mathbf{q}}^{\text{SF}} = \sum_{\mathbf{R}_{\mu}} (\hat{\mathbf{v}} \cdot \hat{\mathbf{z}}_{\mu}) \tau_{\mathbf{R}_{\mu}}^z e^{i\mathbf{q} \cdot \mathbf{R}_{\mu}}$, where $\mathbf{p} \perp \mathbf{q}$ defines a neutron polarization vector and $\mathbf{v} = \mathbf{p} \times \mathbf{q}$ defines the other transverse direction. Therefore, $f_Q(S_{\mathbf{q}}^{\text{DO}}) = f_Q(S_{\mathbf{q}}^{\text{NSF}}) + f_Q(S_{\mathbf{q}}^{\text{SF}})$, since $A^{\text{DO}}(\mathbf{q}) = \langle S_{\mathbf{q}}^{\text{NSF}} S_{\mathbf{q}}^{\text{NSF}} \rangle + \langle S_{\mathbf{q}}^{\text{SF}} S_{\mathbf{q}}^{\text{SF}} \rangle$ by construction. This defines $f_Q(S^{\text{DO}})$ as another sum witness whose individual components are well-bounded via Eq. (11). Thus, $f_Q(S_{\mathbf{q}}^{\text{DO}})$ is bounded by the sum of the underlying NSF and SF channels. Applying Eq. (11) to $S_{\mathbf{q}}^{(\text{NSF})}$, we arrive at

$$f_Q(S_{\mathbf{q}}^{(\text{NSF})}, T) > m/3, f_Q(S_{\mathbf{q}}^{\text{DO}}, T) > 2m/3, \quad (13)$$

which implies that the state is at least $(m+1)$ -partite entangled. See SM (The SM details our multi-directed-loop QMC algorithm for efficient sampling of the QSI model, the ED and the gauge field theory calculation and the computation of neutron scattering cross section in experimental settings.) for a detailed derivation.

Data availability

The data that support the findings of this study are provided at <https://doi.org/10.25442/hku.32404548>.

Code availability

The main codes in this paper are available at <https://doi.org/10.25442/hku.32412273>.

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Author contributions

Z.Y.M. and Y.B.K. supervised the project. C.K.Z. developed the MDL update scheme and performed the QMC simulations. Z.Z. and F.D. performed the ED calculations and the GMFT calculations. All authors contributed to writing the manuscript.

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Competing interests

The authors declare no competing interests.

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